Sophie Nguyen

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Skills

Programming Languages	Python, SQL, Bash
Development Tools	Git, Docker, VS Code
Cloud & Infrastructure	AWS, Terraform
Data & Analytics Tools	Spark, dbt, Airflow, Tableau, Bigquery

Work Experience

Senior Scientific Programmer April 2023 – June 2025
BAYLOR COLLEGE OF MEDICINE, Human Genome Sequencing Center

- Fine-tuned a deep learning variant calling model by augmenting training data from public genomic datasets, achieving a 40% improvement in detecting low-frequency variants.
- Analyzed AWS Cost & Usage Reports to monitor spending, ensure budget compliance, and identify cost anomalies, resulting in $2,000 monthly cloud savings and improved resource allocation efficiency.
- Engineered Dockerized applications and integrated them into serverless workflows, optimizing scalable genomic infrastructure for high-throughput data ingestion, quality control, and variant reporting.
- Used AWS Inspector to assess container security and unintended network exposure, reducing high-risk vulnerabilities by over 90% per monthly cycle and strengthening application security and compliance.
- Benchmarked bioinformatics tools and validated clinical genomic assays to ensure SOP compliance and CAP/CLIA standards. Documented findings and communicated key insights to stakeholders.
- Supported research initiatives through bioinformatics analysis by leveraging statistical methods to extract key insights and deliver actionable reports that informed critical research decisions.

Data Scientist Jan 2020 – Mar 2023
IMMATICS BIOTECHNOLOGIES

- Designed and implemented a graphical user interface with an optimized SQL database, enabling efficient data exploration, insightful visualization, and advanced data mining for complex research initiatives.
- Conducted comprehensive testing of front-end and back-end systems, ensuring robust data integrity, performance, and user experience across all functionalities.
- Integrated proteomics tools with multiple modalities into internal mass spectrometry pipelines, driving the discovery of novel immunotherapeutic HLA peptides.
- Managed reliable execution of bioinformatics pipelines by coordinating scheduling and minimizing downtime through prompt diagnosis and efficient resolution of technical issues during runtime.

Graduate Research Assistant *Georgia Institute of Technology* Aug 2018 – Dec 2019

- Successfully performed exploratory single-cell RNA sequencing analysis on blood samples to investigate gene expression of immune cells, extracting key biological insights, and communicating these effectively.
- Automated scripts to process and analyze large-scale proteomic datasets for mass-spec facilities.

Data Science Intern *Corteva Agriscience* May 2019 – Aug 2019

- Leveraged high-performance computing and web scraping for genomic mining of bioactive natural products, building expression profiles with comparative transcriptomic tools.
- Applied supervised learning to identify biosynthetic gene clusters and their potential functionalities.

Education

2018-2019	**Master of Science, Bioinformatics** at Georgia Institute of Technology
2012-2016	**Bachelor of Arts, Biochemistry** at College of Wooster